THE PRESIDENT'S OFFICE – RALG

TAHOSSA ROMBO DISTRICT EXAMINATION SYNDICATE

FORM FOUR MOCK EXAMINATION

032/2A **CHEMISTRY 2A (ACTUAL PRACTICALS)**

TIME: 3:00 Hours **MARKING SCHEME** **August, 2024**

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a). Table of results (burette readings).

@$1^1/_2$ = Total marks 06 marks

-When pipette used was $25cm^3$

Titration	Pilot	1	2	3
Final burette reading (cm^3)	24.90	23.90	24.20	23.90
Initial burette reading (cm^3)	0.00	0.00	0.00	0.00
Volume used (cm^3)	24.90	23.90	24.20	23.90

-When pipette used was $20cm^3$

Titration	Pilot	1	2	3
Final burette reading (cm^3)	20.00	19.10	19.00	18.90
Initial burette reading (cm^3)	0.00	0.00	0.00	0.00
Volume used (cm^3)	20.00	19.10	19.00	18.90

The volume of pipette used was **20cm3** or **25 cm^3**

Average volume used of solution N (acid) $= \frac{23.9cm^3 + 24.20cm^3 + 23.90cm^3}{3}$ **(01marks)**

$= \frac{73.00cm^3}{3}$

$=$ **$24.00cm^3$**

The volume of pipette used was **$25.00cm^3$** **01marks**

The volume of burette used was **$24.00\ cm^3$** **01marks**

Summary

$24.00\ cm^3$ of solution N was used **$25.00cm^3$** of solution D for complete Neutralization. @01 =**Total 02**

b). The colour change at the End point was from **Yellow** to **Orange.**@01 =**Total 02**

c). The volume of acid for the above neutralization was **$24.00\ cm^3$.01marks**

d). The balanced equation;

$2HCl + T_2CO_3 \longrightarrow 2TCl + H_2O + CO_2$(02marks)

e). To Calculate

i. The Molarity of solution N.

Data Given; Mass of HCl (acid) =1.55g , Volume of HCl (acid) $=200cm^3$

Molarity = $\frac{\text{Conc.g/dm}^3}{\text{Molar mass}}$

Conc,g/dm^3 of an Acid (HCl) = $\frac{\text{Mass (g)}}{\text{Volume (dm}^3\text{)}}$ **01marks**

Where Mass = 1.55g

Volume = 200cm^3 but 1dm^3 = 1000cm^3

1dm^3 = 1000cm^3

? ⟵ 200cm^3

$\frac{1\text{dm}^3 \times 200\text{cm}^3}{1000\text{cm}^3}$ **01marks**

=0.2dm^3

Conc in g/dm^3 of an Acid = $\frac{1.55\text{g}}{0.2\text{dm}^3}$

= 7.75g/dm^3 01marks

Molarity of acid (HCl) = $\frac{\text{Conc in g/dm3}}{\text{Molar mass of HCl}}$ **01marks**

Molar mass of HCl = (1x1) +(35.5 x 1)

= 1 + 35.5

= 36.5g/mol

Molarity = $\frac{7.75\text{g/dm}^3}{36.5\text{g/mol}}$

0.21mol/dm^3

Molarity of an Acid solution (HCl) is 0.21mol/dm^3 01marks

ii. The molarity of Base solution

Applying the formula $\frac{\text{MaVa}}{\text{MbVb}} = \frac{\text{na}}{\text{nb}}$ **01marks**

Where

Ma= Molarity of acid

Mb= Molarity of base

Va= Volume of acid

Vb= Volume of base

na= mole ration of acid

nb= mole ration of base

Data given;

Ma= 0.21M

Mb= Unknown

Va= 24.00cm^3 **01marks**

Vb= 25.00cm^3

na= 2

nb= 1

Molarity of Base(**T_2CO_3**) Mb = $\frac{\text{Ma x Va x nb}}{\text{Vb x na}}$

= $\frac{0.21\text{M} \times 24.00\text{cm}^3 \times 1}{25.00\text{cm}^3 \times 2}$ **01marks**

= **0.1M**

The molarity of Base is 0.1M01marks

iii. To find the Molecular weight of $\mathbf{T_2CO_3}$

Molecular Weight $= \frac{\text{Conc in g/dm}^3}{\text{Molarity}}$

Conc in g/dm3 of $\mathbf{T_2CO_3} = \frac{\text{Mass (g)}}{\text{Volume (dm}^3\text{)}}$ **01marks**

Where Mass= 5.3g, Volume = 0.50dm^3

Conc in g/dm^3 $\mathbf{T_2CO_3}$ $= \frac{5.3\text{g}}{0.50\text{dm}^3}$

$=$ **10.6g/dm^3 (01marks)**

Molecular mass of $\mathbf{T_2CO_3}$ $= \frac{10.6\text{g/dm}^3}{0.1\text{M}}$

$=$ **106g/mol**

The molecular weight of $\mathbf{T_2CO_3}$ is 106g/mol.01marks

iv. To find the atomic mass of element **T** in $\mathbf{T_2CO_3}$

$\mathbf{T_2CO_3}$= Molecular weight

Tx2 + (12 x1) + (16 x 3) = 106.

2T + 12 + 48 = 106

2T+ 60 - 60 = 106 - 60

$\frac{2T}{2} = \frac{46}{2}$

T = 23

Atomic mass of element T is 23.01marks

v. Element T is most Probably **Sodium (Na) 01marks**

2. (a) **@ 01 Total marks 08**

Experiment No	Volume of HH (cm^3)	Volume of TT (cm^3)	Volume of Distilled Water (cm^3)	Time in (sec)	Rate(1/t) (sec^{-1})
1	10	20	0	20	0.05
2	10	15	5	24	0.04
3	10	10	10	34	0.03
4	10	5	15	50	0.02

(b). Balanced equation.

$Na_2S_2O_{3(aq)} + 2HCl_{(aq)} \longrightarrow 2NaCl_{(aq)} + H2O_{(l)} + SO_{2(g)} + S_{(s)}$(02 marks)

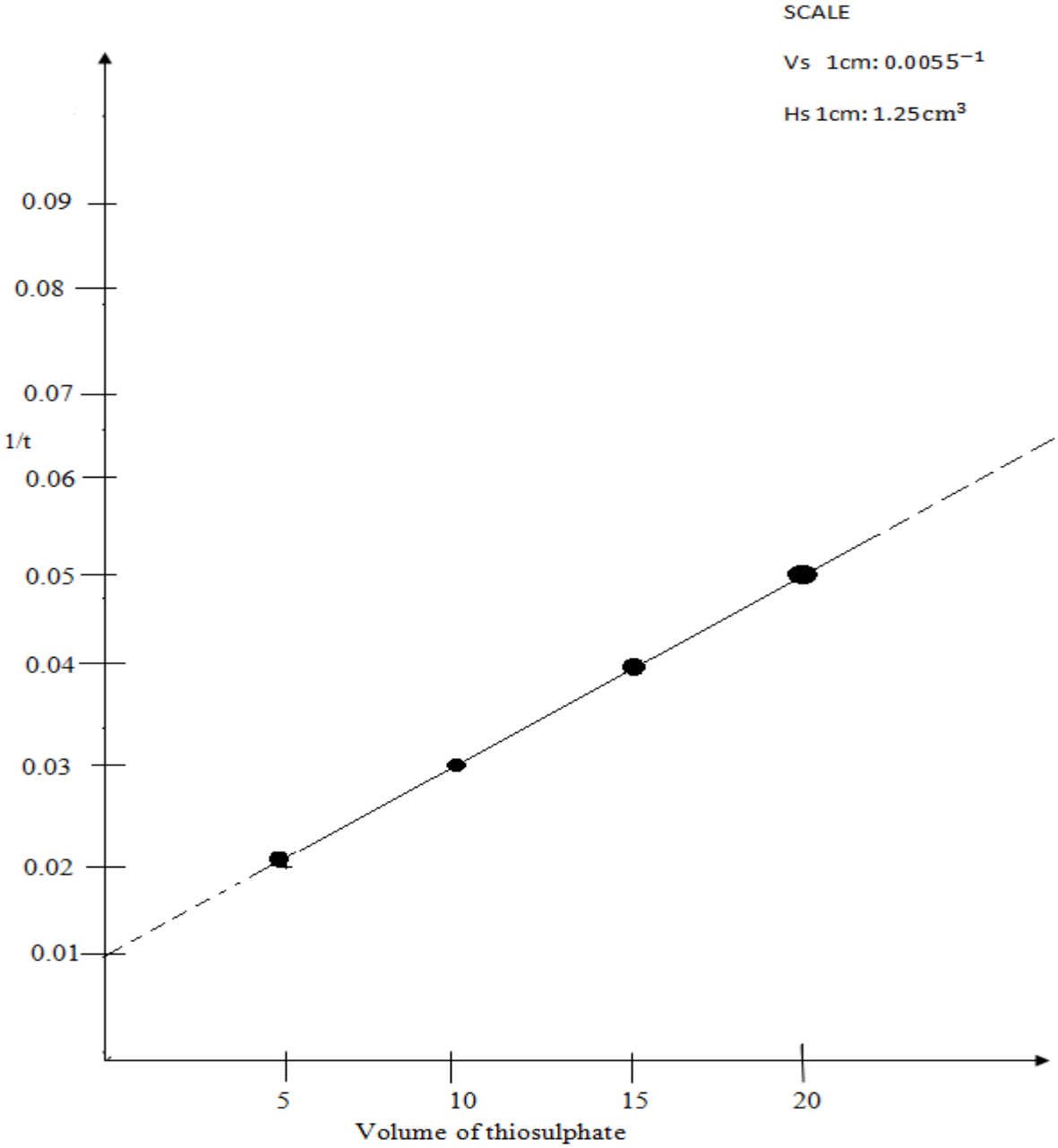
(c). Due to production of Sulphur solid ($S_{(s)}$) **(02 marks)**

(d). The factor of Concentration of Sodium thiosulphate varied to this experiment due to addition of water to

Solution TT will cause to decrease concentration and also rate of reaction will decrease**(02 marks)**

Drawing 01 marks , Sketching 02 marks, heading 01 marks Total **(04 Marks)**

e) A GRAPH OF $\frac{1}{t}$ AGAINST THE VOLUME OF THIOSULPHATE



f) As the concentration of sodium thiosulphate was decreased, the rate of chemical reaction was also decreased and or as the concentration increases, the rate of reaction also increases.

02 marks